

SANAH SURI

Ph.D. in Mathematics

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EDUCATION

Washington University in St. Louis

August 2025

Ph.D. in Mathematics

Thesis: Functional Equivariance and Backward Error Analysis

Advisor: Dr. Ari Stern

Washington University in St. Louis

May 2022

A.M. in Mathematics

Grinnell College

May 2020

B.A. in Mathematics (with honors) & Computer Science

RESEARCH INTERESTS

Machine Learning, Physical Oceanography, Explainable & Interpretable AI, Numerical Analysis

PUBLICATIONS

- **S.Suri**, K. Ringel, and M. Sonnewald (2026) OceanCBM: A Concept Bottleneck Model for Mechanistic Interpretability in Ocean Forecasting, *Accepted as NeurIPS Spotlight Paper*.
- **S. Suri**, and M. Sonnewald (2026), Trusting Machine Learning with Physics: A Fidelity Verification Framework for Neural Networks, *In Press at Journal of Geophysical Research: Machine Learning and Computation*.
- **S. Suri** (2026), Functional Equivariance and Backward Error Analysis, *Ph.D. Thesis*.
- A. Stern, and **S. Suri** (2023), Functional Equivariance and Modified Vector Fields, *Journal of Computational Dynamics*, 11 (4), 409-426.
- D. Rim, **S. Suri**, S. Hong, K. Lee, R. J. LeVeque (2023), A Stability Analysis of Neural Networks and Its Application to Tsunami Early Warning, *Journal of Geophysical Research: Machine Learning and Computation*, 1 (4).
- M. Chamberland, S. Jing, **S. Suri** (2020), A Generalization of the One-Seventh Ellipse, *Mathematics Magazine*, 93:4, 271-275.
- A. Stern, **S. Suri**, Conjugate Functional Equivariance, In Preparation.

WORK EXPERIENCE

Postdoctoral Scholar

Aug 2025-Present

UC Davis, Computational Climate and Ocean Group

Davis, CA

- Developing interpretable machine learning models for ocean dynamics, focusing on validating learned representations of physical processes
- Mentoring graduate students through reading groups and collaborative research projects in climate AI
- Utilizing high-performance computing (HPC) for large-scale model training and analysis of ocean and climate datasets
- Contributing to grant proposals and interdisciplinary research initiatives in computational climate science

Machine Learning Intern

Jun 2022-Aug 2022

*Impossible Sensing**St. Louis, MO*

- Developed machine learning models to predict mineral oxide composition from Raman spectra, outperforming baseline methods
- Curated and preprocessed large-scale spectroscopy datasets from open-source mineral databases
- Aligned model development with deep-ocean sensing objectives; presented results to cross-functional teams

PRESENTATIONS

Institute for Pure and Applied Mathematics (IPAM) at UCLA*February 2026*

Poster on ‘Trusting Machine Learning with Physics’

UC Davis Coastal and Marine Sciences Institute Symposium*October 2025*

Presentation on ‘Trustworthy AI for Ocean Circulation’

WashU Spring Research Roundtable*March 2025*

Presentation on ‘Explainable AI in Ocean Science’

WashU Environmental Research Symposium*February 2025*

Selected to present research on ‘Explainable AI in Ocean Science’

Missouri S&T Applied Math and Statistics Student Seminar*October 2024*

Invited talk on ‘Functional Equivariance and Backward Error Analysis’

SciCADE 2024*July 2024*

Invited talk in minisymposium ‘Recent Advances in Structure Preserving Numerical Methods’

WashU Graduate Research Symposium 2024*March 2024*

Poster Presentation on ‘A Stability Analysis of Neural Networks and Applications to Tsunami Early Warning’

Midwest Numerical Analysis Day*April 2024*

Invited talk in minisymposium ‘Numerical Time Integration’

8th Annual Meeting of SIAM Central States Section*October 2023*

Invited talk on ‘Modified Vector Fields and Functional Equivariance’

Szego Seminar*March 2023*

Expository talk on ‘B-Series’ in a graduate student seminar

Joint Math Meetings (JMM)*January 2020*

Poster presentation on ‘Colorings of Algebraic Structures’

Nebraska Conference for Undergraduate Women in Math (NCUWM)*January 2019*

Talk on ‘The One-Seventh Ellipse Problem’

Joint Math Meetings (JMM)*January 2019*

Poster presentation on ‘The One-Seventh Ellipse Problem’

CONFERENCES & WORKSHOPS

ICERM Hot Topics Workshop: From Modeling to Learning with HPC*September 2025*

Workshop on simulation and machine learning integration for scientific computing

Simons Laufer Mathematical Sciences Institute (formerly MSRI)*Summer 2023*

Graduate summer school on machine learning and topological data analysis at UCSD

Midstates Consortium for Math and Science (MCMS) Undergraduate Research Symposium*November 2022*

Graduate student panelist

7th Annual Meeting of SIAM Central States Section*October 2022*

Attended

SIAM Annual Meeting*July 2022*

Attended

Internship Network in the Mathematical Sciences (INMAS)*October 2021-May 2022*

Workshop spanning academic year 2021-22 training PhD students in Mathematics to develop strong programming, statistics, machine learning and data analysis skills

SAMSI Workshop on Data-Driven Mathematical and Statistical Modeling*Summer 2021*

Intensive workshop held annually on statistical and mathematical modeling using real data culminating in a focused project on Bayesian inference

Midstates Consortium for Math and Science (MCMS) Undergraduate Research Symposium*November 2020*

Graduate Student Panelist

TEACHING EXPERIENCE

Assistant to the Instructor

May 2021 - May 2024

*Washington University in St. Louis**St. Louis, MO*

- Spring 2024: Calculus 3
- Fall 2023: Calculus 3
- Spring 2023: Calculus 1
- Fall 2022: Calculus 1
- Spring 2022: Calculus 3
- Fall 2021: Calculus 3
- Summer 2021: Calculus 2
- Tutor in Mathematics and Statistics Help Room

Directed Reading Program

January 2023 - May 2024

*Washington University in St. Louis**St. Louis, MO*

- Spring 2024: Mathematical Machine Learning
- Spring 2023: Introduction to B-Series

HONORS & AWARDS

Co-Investigator, NAIRR Pilot Start-Up Project

March 2026

Second Place, WashU Graduate Research Symposium

March 2024

Brian Blank Award for Outstanding Third Year Graduate Student

May 2023

Outstanding Poster at Joint Math Meetings

January 2020

Third Place, Iowa Intercollegiate Math Competition

*Spring 2018***LEADERSHIP & SERVICE**

President and Founder, SIAM Student Chapter at WashU

Fall 2023 - May 2025

Department Representative, Arts & Sciences Graduate Student Association

August 2023 - May 2024

Member, Diversity, Equity and Inclusion Committee

August 2022 - May 2023

Peer Mentor, WashU Department of Mathematics

*August 2022 - May 2023***PROFESSIONAL MEMBERSHIPS**

Association for Women in Mathematics (AWM)

American Mathematical Society (AMS)

Society for Industrial and Applied Mathematics (SIAM)